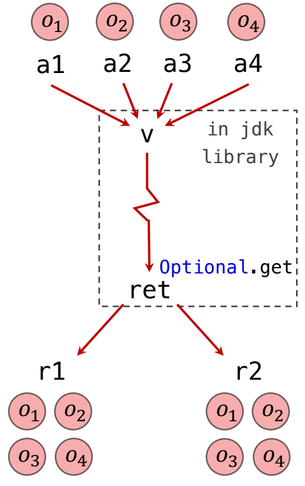
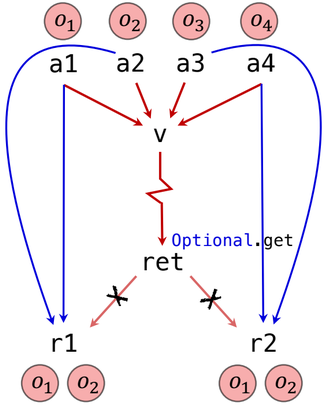


(a) Dynamic behavior of the first pipeline instance.



(b) PFG fragment under context insensitivity



(c) PFG fragment under CUT-SHORTCUT^S

```
1  Stud a1 = new Stud("1", "Alice"); // o1
2  Stud a2 = new Stud("2", "Bob"); // o2
3  Stud a3 = new Stud("3", "Charlie"); // o3
4  Stud a4 = new Stud("4", "Dean"); // o4
5  Stream t1 = Stream.of(a1, a2); // h1
6  Function f1 = Stud::getName; // λ1
7  Stream t2 = t1.map(f1); // h1'
8  Optional t3 = t2.findFirst();
9  String r1 = t3.get();
10 Stream t4 = Stream.of(a3, a4); // h2
11 Function f2 = Stud::getID; // λ2
12 Stream t5 = t4.map(f2); // h2'
13 Optional t6 = t5.findFirst();
14 String r2 = t6.get();
```

StruCre (5) StruPro (8) StruOut (9) StruCre (10) FuncPro (12) StruPro (13) StruOut (14)

Target(λ_1)

```
String getName() {
  ret = this.name;
  return ret;
}
```

Target(λ_2)

```
String getID() {
  ret = this.id;
  return ret;
}
```

```
class ReferencePipeline{
  ...
  map(Function mapper){
    ...
    t = mapper.apply(e);
    ... lambda callsite l
  }
}
```

Library Code

call call call call